

Common Childhood Injuries & First Aid

Burns, Cuts, Scrapes, Sprains, Insect Bites, Bumps, Bruises, Splinters, and Sunburn

Based on AAP, Red Cross, Mayo Clinic, and CDC guidelines. For educational purposes only.

1. Burns in Children

Burns are classified by depth. Children have thinner skin than adults, so burns can be more severe and deeper than they appear initially.

Types of Burns

- **First-degree burn (superficial):** Affects only the outer layer of skin (epidermis). Red, dry, painful, no blisters. Like a mild sunburn. Heals in 3–5 days without scarring.
- **Second-degree burn (partial thickness):** Affects the epidermis and part of the dermis. Red, blistered, very painful, moist/weeping appearance. Heals in 2–3 weeks. May scar. Second-degree burns larger than 3 inches or on the face, hands, feet, groin, or over a joint need medical attention.
- **Third-degree burn (full thickness):** Destroys the full thickness of skin. May appear white, brown, or charred. May be painless because nerve endings are destroyed. Always requires emergency medical care. Skin grafting may be needed.

Common Causes in Children

- Scalds from hot liquids (most common in children under 5) — hot coffee, tea, soup, bathwater, steam
- Contact burns from touching hot surfaces — stoves, ovens, curling irons, space heaters, fireplaces
- Flame burns from fire, candles, lighters, matches
- Electrical burns from outlets, cords, appliances (always need medical evaluation even if skin looks fine)
- Chemical burns from cleaning products, drain cleaners, battery acid
- Sunburn from UV exposure

First Aid for Minor Burns (First-Degree and Small Second-Degree)

HOME CARE: Most minor burns can be treated at home.

- **Step 1 — Cool the burn immediately.** Run cool (not ice cold) water over the burned area for 10–20 minutes. This is the single most important first aid step. Do this within the first few minutes for best results.
- **Step 2 — Remove clothing and jewelry** from the burned area, unless stuck to the skin. If clothing is stuck, do not pull it off — cut around it.
- **Step 3 — Protect the burn.** After cooling, cover loosely with a clean, non-stick bandage or sterile gauze. Do not wrap tightly.
- **Step 4 — Pain relief.** Acetaminophen or ibuprofen (age-appropriate dosing) for pain. Cool compresses can also help.
- **Step 5 — Moisturize.** Once cooled, apply pure aloe vera gel or a gentle moisturizer. After the first 24 hours, you can apply a thin layer of antibiotic ointment (like bacitracin) and re-cover with a non-stick bandage. Change the bandage daily.
- **Step 6 — Watch for infection.** Signs of infection: increasing redness, swelling, warmth, pus or cloudy drainage, red streaks radiating from the burn, fever, or increasing pain after the first 24–48 hours.

What NOT to Do

- Do NOT apply ice or ice water (causes further tissue damage and can worsen the burn)
- Do NOT apply butter, oil, toothpaste, or any home remedies (traps heat, increases infection risk)
- Do NOT pop blisters (blisters protect the healing skin underneath; popping increases infection risk)
- Do NOT use adhesive bandages directly on a burn (use non-stick gauze instead)
- Do NOT apply antibiotic ointment to third-degree burns or burns with open wounds without medical guidance

Blister Care

- Leave blisters intact — they are the body's natural bandage protecting the new skin forming underneath.
- If a blister breaks on its own: gently clean with mild soap and water, apply antibiotic ointment, and cover with a non-stick bandage.
- Do not peel off the loose skin from a broken blister — it provides protection while healing.

CALL 911 / GO TO ER IF: Burn is on the face, hands, feet, genitals, or over a major joint. Burn is larger than 3 inches (about the size of the child's palm). Burn goes all the way around an arm, leg, finger, or toe (circumferential burn). Burn is third-degree (white, brown, or charred; painless). Burn was caused by electricity or chemicals (always need medical evaluation). Child is under 1 year old with any burn beyond mild sunburn. Signs of infection develop. Child has difficulty breathing (possible airway burn from inhaling hot air/steam). Burns with blisters in a child under 2 years old.

Burn Prevention

- Set water heater to 120°F (49°C) or below. Test bathwater with your elbow before putting a child in.
- Keep hot liquids away from table edges. Turn pot handles toward the back of the stove. Use back burners when possible.
- Keep children away from stoves, ovens, fireplaces, space heaters, curling irons, and candles.
- Cover electrical outlets. Keep cords out of reach. Never let children play with electrical appliances.
- Apply sunscreen SPF 30+ to children over 6 months before sun exposure. Reapply every 2 hours. Keep infants under 6 months out of direct sunlight.
- Install smoke detectors on every level of the home. Practice a fire escape plan.
- Store matches, lighters, and flammable materials locked away and out of reach.

Sources: AAP HealthyChildren.org; Mayo Clinic Burns First Aid; Red Cross First Aid Guidelines; CDC Burn Prevention

2. Cuts and Scrapes

Minor cuts (lacerations) and scrapes (abrasions) are among the most common childhood injuries. Most can be safely treated at home.

Home Treatment

- **Step 1 — Stop the bleeding.** Apply gentle, firm pressure with a clean cloth or gauze for 5–10 minutes. Do not keep lifting to check — this disrupts clot formation. Most minor cuts stop bleeding within 10 minutes.
- **Step 2 — Clean the wound.** Rinse under clean running water for 5 minutes to remove dirt and debris. Use mild soap around (not directly in) the wound. Do not use hydrogen peroxide or rubbing alcohol — these damage healthy tissue and slow healing.
- **Step 3 — Apply antibiotic ointment.** A thin layer of bacitracin or Neosporin helps prevent infection and keeps the wound moist for faster healing.
- **Step 4 — Cover the wound.** Use an adhesive bandage or sterile gauze. Change the bandage daily and whenever it gets wet or dirty. Keep the wound covered until it has fully closed.
- **Step 5 — Watch for infection.** Signs: increasing redness, swelling, warmth, pus, red streaks, fever, or increasing pain after 24–48 hours.
- **Tetanus:** Make sure your child's tetanus vaccination is up to date (DTaP series for young children, Tdap booster for older children). If the wound is deep or contaminated with dirt and the child hasn't had a tetanus shot in 5+ years, contact your pediatrician.

CALL 911 / GO TO ER IF: Bleeding does not stop after 10–15 minutes of firm pressure. Cut is deep (you can see fat or muscle tissue), gaping, or has jagged edges. Cut is on the face (may need careful closure to minimize scarring). Cut was caused by a dirty or rusty object. Cut was caused by an animal or human bite. Signs of infection develop. Object is embedded in the wound (do not remove it — stabilize it and go to the ER). Cut is longer than half an inch and edges do not stay together — likely needs stitches. Stitches need to be applied within 6–12 hours for best results (facial wounds within 24 hours).

Sources: AAP; Mayo Clinic; Red Cross

3. Sprains and Strains

A sprain is a stretch or tear of a ligament (connects bone to bone). A strain is a stretch or tear of a muscle or tendon (connects muscle to bone). Ankle and wrist sprains are the most common in children. True sprains are uncommon in very young children because their growth plates are weaker than ligaments — what looks like a sprain in a young child may actually be a fracture.

Home Treatment (RICE Method)

HOME CARE: Use RICE for the first 48–72 hours:

- **R — Rest:** Keep weight off the injured area. Use crutches if needed for a leg/ankle injury.
- **I — Ice:** Apply an ice pack wrapped in a thin towel for 15–20 minutes every 2–3 hours. Never apply ice directly to skin. For young children, 10 minutes is sufficient.
- **C — Compression:** Wrap the injured area with an elastic bandage (ACE wrap) snugly but not tightly. If the child complains of numbness, tingling, or the area turns blue, loosen the wrap.
- **E — Elevation:** Raise the injured limb above the level of the heart when possible to reduce swelling.
- **Pain relief:** Ibuprofen (for children 6 months+) is preferred because it reduces both pain and inflammation. Acetaminophen also works for pain.

CALL 911 / GO TO ER IF: Inability to bear weight on the injured leg or use the injured arm. Visible deformity (bone looks out of place or bent). Severe swelling that gets worse rapidly. Numbness or tingling below the injury. Popping sound at the time of injury. Child is under 5 years with a suspected sprain (may be a fracture — growth plates are vulnerable). Pain and swelling do not improve after 48–72 hours of RICE.

Sources: AAP; Mayo Clinic; Red Cross

4. Insect Bites and Stings

- **Bee/wasp stings:** Remove the stinger by scraping sideways with a credit card or fingernail (do not squeeze with tweezers — this injects more venom). Wash with soap and water. Apply ice for 10 minutes. Apply hydrocortisone cream or calamine lotion for itch. Give oral antihistamine (cetirizine/Zyrtec or diphenhydramine/Benadryl) if swelling or itching is bothersome.
- **Mosquito bites:** Wash with soap and water. Apply calamine lotion or hydrocortisone cream. Avoid scratching (can cause infection). Trim fingernails to reduce scratching damage. An ice cube can reduce itch.
- **Tick bites:** Remove the tick with fine-tipped tweezers by grasping as close to the skin as possible and pulling straight up with steady, even pressure. Do not twist, crush, or burn the tick. Clean the area with soap and water. Save the tick in a sealed bag (your doctor may want to identify it). Watch for a bulls-eye rash, fever, or body aches in the next 30 days — these may indicate Lyme disease.

CALL 911 / GO TO ER IF: Difficulty breathing, throat swelling, or widespread hives after a sting (anaphylaxis — use EpiPen if available and call 911 immediately). Swelling that spreads significantly beyond the sting site. Fever, body aches, or bulls-eye rash after a tick bite. Signs of infection at the bite site (increasing redness, warmth, pus).

Sources: AAP; CDC Tick Prevention; Mayo Clinic

5. Bumps, Bruises, and Head Injuries

Bruises (contusions) are caused by bleeding under the skin from a bump or fall. They change color as they heal: red/pink → blue/purple → green/yellow → brown → fading. Most bruises on the arms, legs, and forehead are harmless and heal in 1–2 weeks.

Home Treatment for Bruises

- Apply a cold compress or ice pack (wrapped in a towel) for 10–20 minutes to reduce swelling
- Elevate the injured area if possible
- Acetaminophen or ibuprofen for pain (avoid aspirin)

Head Bumps in Children

Head bumps are common in toddlers and young children. Most are minor (a bump on the forehead that swells into a "goose egg"). However, head injuries require careful monitoring.

HOME CARE: After a head bump, monitor the child closely for 24–48 hours.

- Apply ice wrapped in a cloth to the bump for 20 minutes
- Let the child rest but do not prevent them from sleeping — the old advice to "keep them awake" is outdated. It is OK to let the child sleep, but check on them periodically
- Give acetaminophen for pain (avoid ibuprofen for the first 24 hours after head injury as it can increase bleeding risk)

CALL 911 / GO TO ER IF: Loss of consciousness (even briefly). Vomiting more than once after the injury. Confused, unusually drowsy, or difficult to wake. Unequal pupils (one bigger than the other). Severe or worsening headache. Clear fluid draining from nose or ears. Seizure. Weakness or numbness in arms or legs. The child is under 6 months. Difficulty walking, speaking, or seeing. Behavior changes (more irritable, less responsive than usual). Bump is on the side or back of the head (higher risk than forehead bumps). Fall from a significant height (more than 3 feet for children under 2, more than 5 feet for older children).

Sources: AAP; CDC Heads Up Concussion; Mayo Clinic

6. Splinters

- **Small, visible splinters:** Clean the area with soap and water. Sterilize tweezers with rubbing alcohol. Grasp the splinter as close to the skin as possible and pull in the direction it entered. Clean again and apply antibiotic ointment with a bandage.
- **Splinter just under the surface:** Sterilize a needle with rubbing alcohol. Gently break the skin over the end of the splinter and lift it out. Then use tweezers to pull it the rest of the way.
- **Deep or difficult splinters:** If you cannot remove it, soak the area in warm soapy water for 10–15 minutes twice daily — this may work the splinter to the surface naturally over 1–2 days.

CALL 911 / GO TO ER IF: Splinter is deeply embedded, very large, or under a fingernail. Splinter is glass, metal, or from a cactus/sea urchin. Signs of infection develop. Child is not up to date on tetanus vaccination.

Sources: AAP; Mayo Clinic

7. Sunburn

Children's skin is thinner and more sensitive to UV radiation than adult skin. Sunburns in childhood significantly increase the risk of skin cancer later in life. Infants under 6 months should be kept out of direct sunlight entirely.

Home Treatment

- **Cool baths or compresses:** Apply cool (not cold) damp cloths to the burned areas for 10–15 minutes several times a day.
- **Moisturize:** Apply pure aloe vera gel or a gentle, fragrance-free moisturizer while the skin is still damp. Avoid products containing benzocaine or lidocaine (can cause allergic reactions).
- **Hydration:** Sunburn draws fluid to the skin surface and away from the body. Offer extra fluids for 2–3 days.
- **Pain relief:** Ibuprofen to reduce pain and inflammation. Acetaminophen also works.
- **Do not pop blisters.** If blisters break, clean gently and apply antibiotic ointment.
- **Loose, soft clothing** over sunburned areas. Keep out of the sun until fully healed.
- **Prevention:** SPF 30+ broad-spectrum sunscreen on children over 6 months, reapply every 2 hours and after swimming/sweating. Protective clothing, wide-brimmed hats, and shade during peak hours (10 AM–4 PM). For infants under 6 months: shade and protective clothing only, no sunscreen.

CALL 911 / GO TO ER IF: Sunburn with blisters covering a large area. Fever, chills, headache, or confusion after sunburn (possible sun poisoning/heat illness). Sunburn in an infant under 1 year. Signs of dehydration.

Sources: AAP; CDC Sun Safety; Mayo Clinic

Disclaimer: Compiled for the PediatricAI project based on AAP, Red Cross, CDC, and Mayo Clinic guidelines. NOT a substitute for professional medical advice.